

PROJECT PROPOSAL: PEFT BAKE-OFF

Advanced NLP Group | Efficiency Benchmarking Project

This document outlines the systematic comparison of Parameter-Efficient Fine-Tuning (PEFT) methods on small-to-medium Large Language Models (LLMs). The goal is to evaluate trade-offs between trainable parameters, memory efficiency, and mathematical reasoning performance.

1. Target Models & Dataset

- **Base Models:** Qwen-2.5-1.5B (Base) and Llama-3.2-1B (Base). We use base versions to observe raw learning deltas without the influence of prior instruction tuning.
- **Dataset:** **GSM8K**. This math word problem dataset provides a clear metric (accuracy) for evaluating reasoning capabilities under low-rank constraints.

2. Implementation Scope (The "Bake-Off")

To satisfy course requirements, we will implement custom layers and surgery scripts for the following methods:

Method	Implementation Strategy	Focus Metric
Full FT	Standard baseline (A6000 Cluster).	Performance Upper Bound
Custom LoRA	<code>nn.Linear</code> wrapper; rank $r=16$; target Q/V.	Rank vs. Quality
Custom IA3	Activation scaling vectors (K/V/MLP).	Parameter Minimalism
Prefix Tuning	Learned KV-cache prefixes in Transformer blocks.	Architectural Impact
QLoRA	4-bit quantization via <code>bitsandbytes</code> + LoRA.	Memory Bottleneck

3. Evaluation & Research Goals

- **Efficiency:** Track Peak VRAM and Training Wall-clock time via Weights & Biases (W&B).
- **Ablation:** Study the effect of rank ($r=8$ to 128) and target modules (Attention-only vs. All-Linear).
- **Inference:** Analyze the latency overhead introduced by Prefix Tuning compared to merged LoRA weights.

Next Steps:

1. Finalize the "Surgery Script" to swap `nn.Linear` with `CustomLoRALinear`.
2. Setup W&B logging for system resource monitoring.
3. Run the Full Fine-Tuning baseline to establish the ceiling.